

The Arrow's Horizon: The Khronon's Cosmological Face — the Dark Fluid, the Clustering Dial, and One Cosmological Constant

Allen Proxmire

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Abstract

KM-I unified Event Density's gravity: one khronometric theory — Einstein at high accelerations, MOND below a_0 — with the galactic phenomenology carried by the khronon, the substrate's arrow of time made dynamical. That left the cosmological half: the deep-vacuum soft spot, the standing MOND-class debts (clusters, the CMB), and a flagged two-faces question about Λ . This paper resolves the structure of all three. **First, the state space:** the configuration at which the deep-infrared sector degenerates, $(A, \theta) = (0, 0)$, is exact Minkowski vacuum — **and ED's substrate forbids exact Minkowski vacuum: the tick never halts.** The degeneracy is diagnosed as an artifact of truncating the foliation EFT to the acceleration alone; the general sector $\mathcal{W}(A^2, \Theta)$ ($\Theta = \theta/3H_0$) is healthy at the physical cosmological point, and its Θ -dependence — the **regulator family** — is honestly unfixed. **Second, the orthogonality theorem:** static congruences have $\theta \equiv 0$, so every Θ -term vanishes identically in statics — *the entire tested sector of GR-I/KM-I (the modified Poisson equation, the lensing pass, the screening, the Cassini constraints) is invariant under the regulator choice.* The family is sequestered: no leak-back, no protection. **Third, the cosmological face itself:** on FRW the khronon's Θ -sector is a **dark fluid** $\rho_{\mathcal{W}}(H)$ with an automatic Λ -order late-time constant; and the condition for the dust-like clustering the CMB demands ($c_s \approx 0$) is **exactly the near-vacuum behavior previously flagged as the soft spot** — *the degeneracy cure and the clusters/CMB debt are the same dial*, which the regulator member must set while threading caustics, Čerenkov, BBN, and FRW stability. The class realizes both ends of the dial (mimetic dust at $c_s = 0$; integration-constant dust in the neighboring projectable branch), and the abundance-as-initial-condition character of those mechanisms is consonant with ED's value-inheritance. **Fourth, one Λ :** the khronon's vacuum term and the corpus's V1-backreaction Λ (Paper_038.5) carry the *same scale*, $H_0^2 c^2 / G$, by each paper's own statement — automatically, because both are projections of the one SCBU boundary $R_H = c/H_0$ — and the level diagnosis (substrate mechanism vs EFT constant) converts double-counting into a category error: **one Λ , two descriptions.** The matching is then **performed to its honest limit:** the scaling identity is exact and parameter-free; the sign check **passes** (the V1 vacuum density is positive-definite, giving the observed de Sitter sign and fixing $\mathcal{W}_0 < 0$); and the coefficient is transcribed — $\mathcal{W}_0 = -24\pi^2 \Omega_\Lambda \approx -162$ against a_0^2 , an order-unity $6\Omega_\Lambda \approx 4.1$ against the raw background scale $(cH_0)^2$ — at the inherited tier, so the khronon EFT **loses its last free cosmological constant**, and the ab-initio substrate integral remains open with a *numeric target to miss.* The khronon's portfolio closes into one shape: **one field, one scale, five roles** — the arrow, the Einstein sector, the MOND sector, the dark-fluid

dial, and the Λ -order vacuum term. Clusters and the CMB remain owed until the dial is set and tested; this paper reduces them from an inherited shortfall to one function under five filters, with the data as judge.

1. Introduction

1.1 What this paper does

KM-I ended with the galactic half of the unification derived and passing, and three cosmological items open: the deep-vacuum degeneracy (the one soft spot of the stability analysis), the inherited MOND-class debts (clusters, the CMB), and a flagged reconciliation between two Λ -mechanisms in the corpus. This paper supplies the structure of the cosmological half in four steps: the state-space resolution of the degeneracy (§3), the orthogonality theorem that sequesters the cosmological sector from everything already tested (§4), the FRW analysis — the dark fluid and the clustering dial (§5–§6), and the one- Λ thesis (§7).

Three orienting statements, up front:

- **No derivational hinge is exercised here.** KM-I’s single hinge (the deep-IR matching to Paper_030) is inherited; this paper’s results are a state-space observation, a kinematic theorem, a structural mapping, and a level identification. Where something is a family, it is labeled a family; where a computation is open, it is named.
- **The headline results:** (i) ED’s substrate forbids the configuration at which the theory would degenerate (*no Minkowski vacuum — the tick never halts*); (ii) the cosmological sector is **provably invisible** to every test the companion papers passed (the Orthogonality Theorem); (iii) the dust-clustering condition the CMB demands coincides with the flagged soft spot — **the degeneracy cure and the clusters/CMB debt are one dial**; (iv) the corpus has **one Λ , not two** — the two Λ faces are one constant *at two levels* (the substrate mechanism and its EFT encoding), with the matching performed: scaling exact, sign passing, and the EFT’s vacuum constant **pinned** ($\mathcal{W}_0 = -24\pi^2\Omega_\Lambda$, at the inherited tier).
- **The cosmological sector is a family — the regulator family — and KM-II’s job is to specify its structure and constraints, not to choose a member.** Accordingly, the reader should *not* expect a chosen member, a computed CMB spectrum, a derived Λ value, or a discharged cluster debt. The paper’s product is the *exact specification* of the remaining model-building task: one function, five filters, data as judge.

1.2 Why this matters

Clusters and the CMB are where MOND-class theories go to die: galactic dynamics they handle, but the acoustic-peak structure wants something that clusters like cold dust at recombination, and the cluster masses want more than the modified force law supplies. Most relativistic MOND constructions face this with nothing to offer but epicycles. The khronon construction is differently positioned, for a structural reason this paper makes precise: **the same field already carries a cosmological sector** — not added for the purpose, but forced into existence by the EFT generality of the foliation sector and anchored, like everything else about the khronon, to the one scale H_0 . Whether that sector *delivers* is a model-building and observational question; that

it exists, is sequestered from the tested physics, and sits exactly at the dust-clustering edge is structure, and it is this paper's content.

1.3 How this fits the arc

- **Paper GR-I:** the bandwidth metric and lapse; the weak-field Einstein tests. (*inherited*)
- **Paper GR-II:** the khronometric class; the khronon; GW and Lorentz safety. (*inherited*)
- **Paper KM-I:** the galactic unification — the khronon's deep field as the MOND sector. (*inherited; this paper is its cosmological companion*)
- **Paper_038.5:** Λ as the V1 vacuum integral over $R_H = c/H_0$. (*read from source; unified with, not displaced*)
- **SCBU:** the one boundary $R_H = c/H_0$. (*here acquiring its second projection — and its dynamical carrier's vacuum term*)

1.4 Conventions and regime of validity

Signature $(-, +, +, +)$; $c = 1$ where convenient; $\Theta \equiv \theta/3H_0$, $A^2 \equiv a_\mu a^\mu/a_0^2$, $x \equiv |\nabla\Phi|/a_0$ as in KM-I. The static results referenced are KM-I's (weak-field, quasi-spherical, DCGT limit); the cosmological statements here are **FRW-background structural results** — fluid content, scaling behaviors, and requirement maps, **not** computed spectra or fits. Exact coefficients are convention-dependent and carried schematically.

2. Primitive Inputs

Substrate primitives (postulated, Paper_087): P11/P13 (the arrow and the tick — the khronon's origin and, in §3, the no-Minkowski-vacuum observation), P04 (bandwidth — metric and lapse via GR-I), P02/P05 (adjacency/transport).

Inherited results (load-bearing): GR-I (metric, lapse, Einstein branch); GR-II (khronometric class, the khronon, cosmic-frame anchoring, $c_T = c$); KM-I (the foliation action with the acceleration sector $W(A^2)$, the deep-IR matching, the lensing/screening passes, the IR Cancellation Requirement); Paper_029 ($a_0 = cH_0/2\pi$); Paper_038.5 (Λ as the V1 boundary integral; M3 tier; the value $\rho_\Lambda = (3/8\pi)H_0^2 M_P^2$).

External dictionary (inherited, labeled I): khronometric/aether FRW cosmology (the $\lambda\theta^2$ sector renormalizing G_{cosmo} ; BBN bound); **mimetic dark matter** (a unit-timelike-gradient scalar contributing exact pressureless dust, $c_s = 0$; caustics; the imperfect-fluid refinements); **dark matter as an integration constant** (projectable Hořava — the *neighboring* branch); the classical requirement that the CMB's acoustic structure be fit with a dust-like clustering component.

No new primitive is introduced; no primitive forcing is invoked. The paper exercises no derivational hinge of its own (§1.1).

2.5 Load-Bearing Step Audit

Status legend: **I** = inherited (cited paper or established external result); **D** = derived here; **D-structural** = derived from structure alone; **D-via-I** = derived given an inherited input (the tier transports — fixed, but at inheritance grade); **family** = honestly-unfixed functional freedom; **form-tier** = order-of-magnitude/structural consistency, no value claimed; **thesis** = a structurally-indicated identification; **open** = declared open.

Step	Status	Source / justification
Khronon, class, anchoring; galactic unification	I	GR-II; KM-I
Deep-IR sector degenerates at $(A, \theta) = (0, 0)$	I	KM-I §6/§7 (the flagged soft spot)
$(A, \theta) = (0, 0) = \text{exact}$ Minkowski vacuum, outside ED's state space	D-structural	§3.1 — P11/P13: the tick never halts; the khronon's background is the Hubble flow
Pure- A^2 form is a truncation; $\mathcal{W}(A^2, \Theta)$ is the general same-order sector	D (EFT generality)	§3.2
Generic $\mathcal{W}$ healthy at the physical point $(0, \Theta \sim 1)$	D (EFT generality)	§3.2
The Θ -dependence of $\mathcal{W}$	family	§3.2 — the regulator family, labeled
Orthogonality theorem: $\theta \equiv 0$ in statics $\implies$ tested sector invariant under the regulator	D	§4 — kinematics; the paper's cleanest result
FRW: $\lambda\theta^2 \sim \lambda H^2$ renormalizes G_{cosmo} (BBN-bounded)	I	§5 — class literature
FRW: $\mathcal{W}(0, \Theta)$ is a dark fluid $\rho_{\mathcal{W}}(H)$	D-structural	§5
Late-time constant $\sim a_0^2 c^2 / G \sim H_0^2 c^2 / G$ automatic	form-tier	§5 — the single-scale argument (KM-I §7.4)
Requirements map (BBN / CMB-dust / clusters / late- Λ / Čerenkov+stability)	D (mapping)	§5
Dust-clustering condition $c_s \approx 0$ = the flagged near-vacuum behavior	D-structural	§6 — the soft spot is the doorway ; the central result
The three-way trade (clustering vs caustics vs Čerenkov); the window	D (mapping)	§6
Mimetic dust ($c_s = 0$ end); integration-constant dust (projectable branch)	I — directions	§6 — proofs-of-concept; the latter is the <i>neighboring</i> branch

Step	Status	Source / justification
Abundance as initial condition $\leftrightarrow$ value-inheritance	consonance noted	§6
Both Λ faces carry $H_0^2 c^2 / G$	documentary	§7 — by each paper’s own statement
Scale agreement automatic (shared SCBU anchor)	D-structural	§7 — two projections of one boundary
$\mathcal{W}_0 = \text{EFT encoding of the V1 integral (one } \Lambda)$	thesis	§7 — the level identification
Matching condition (Friedmann-level equality; no double-counting)	D	§7.1 — the EFT constant is <i>defined</i> by the substrate computation
Scaling identity exact ($H_0^2 c^2 / G$ both faces, parameter-free)	D	§7.1
Sign check: V1 density positive-definite $\implies \Lambda > 0 \implies \mathcal{W}_0 < 0$	D (density-level)	§7.1 — coupling-chain confirmation assigned to the ab-initio task
$\mathcal{W}_0 = -24\pi^2 \Omega_\Lambda \approx -162 (= 6\Omega_\Lambda \text{ against } (cH_0)^2)$	D-via-I	§7.1 — tier transported from 038.5, not upgraded
Ab-initio V1 integral (with the §7.1 numeric target)	open	§7.1 — Route-A-class frontier
Matter-sector CC problem	untouched	§7 — stays where 038.5 left it
A regulator member chosen; a spectrum computed; clusters/CMB discharged	NOT DONE / NOT CLAIMED	preamble 2, 6

3. The State Space and the Regulator Family

3.1 ED has no Minkowski vacuum

KM-I’s one soft spot was the deep-vacuum degeneracy: the acceleration sector’s stiffness vanishes as $A \rightarrow 0$. The configuration at which the structure actually fails is $(A, \theta) = (0, 0)$ — zero acceleration *and* zero expansion: **exact Minkowski vacuum, eternal and inert. And ED’s substrate forbids exact Minkowski vacuum: the tick never halts.** The khronon’s background is the Hubble flow (GR-II); commitment does not cease and the tick does not stop (P11, P13). The physical background manifold is the (A, θ) half-space with $\theta \sim H > 0$; the degenerate point sits on its excluded boundary. This is a *domain* observation — nothing about the action is modified — and it is one of the construction’s strongest structural facts: *the substrate’s refusal to ever stop ticking is what protects its gravity theory.*

3.2 The truncation diagnosis and the family

The remaining question is whether perturbations around the *physical* near-vacuum ($A \rightarrow 0, \theta = 3H$) are healthy — and that exposes the deeper point: carrying the deep-IR sector as $W(A^2)$

alone was a **truncation**. The foliation EFT’s invariants at this order are the congruence scalars — acceleration, expansion, shear — and nothing singles out the acceleration as the sole argument of the non-analytic infrared structure. The general same-order sector is

$$W(A^2) \longrightarrow \mathcal{W}(A^2, \Theta), \quad \Theta \equiv \frac{\theta}{3H_0},$$

and at the physical cosmological point $(A, \Theta) = (0, \sim 1)$ the generic member of this class sits at a **generic point of its domain**: there is no structural reason for its quadratic forms to vanish there, so the expansion sector supplies the stiffness the acceleration sector loses. The degeneracy is thereby diagnosed: **an artifact of the truncation, evaluated at a point ED never visits**.

The honest label, as always: the Θ -dependence is **not forced** — not by the substrate, not by the KM-I matching (which constrains only the static/ A sector). It is the **regulator family**, and the next section shows why no member of it needs choosing for anything so far tested.

4. The Orthogonality Theorem

KM-I derived that static congruences have $\theta = \sigma = 0$ identically. Therefore:

Every Θ -dependent term in $\mathcal{W}$ vanishes identically in the static sector.

The entire tested body of the unification — the modified Poisson equation, the GR limit, the Combination-Rule and BTFR embedding, the lensing pass, the AQUAL ellipticity, the screening, and the Cassini constraints — is evaluated at Θ ’s static value and is **invariant under the choice of regulator**.

Both directions of the consequence matter:

- **No leak-back.** The regulator cannot be tuned to assist the galactic physics — it is invisible there. Whatever member the cosmological data eventually select, KM-I’s results stand unchanged. The retrofit worry is not resisted but *structurally impossible*.
- **No protection.** Equally, the tested sector cannot vouch for the regulator: the cosmological sector faces its own filters (§5–§6) on its own. Two independently falsifiable sectors — the healthiest structure available.

5. The Cosmological Face: the Dark Fluid

With the static sector sequestered (§4), we now turn to the FRW background — where the khronon’s Θ -sector stops being invisible and becomes the physics. On FRW the aligned khronon is the comoving congruence ($a = 0$, $\sigma = 0$, $\theta = 3H$), and the foliation sector’s cosmological roles are:

1. $\lambda\theta^2 \sim \lambda H^2$ — a renormalization of the effective cosmological Newton constant; the known khronometric effect, bounded by BBN’s constraint on G_{cosmo}/G_N . [I.] Inherited; nothing new.

2. $\mathcal{W}(0, \Theta)$ — **a dark fluid**. A function of $\Theta = H/H_0$ enters the Friedmann equation as a fluid $\rho_{\mathcal{W}}(H)$ with two automatic features: at late times ($H \rightarrow H_0$) it tends to a **constant of order** $a_0^2 c^2/G \sim H_0^2 c^2/G$ — the Λ -order vacuum term (form-tier; §7) — and at early times its behavior is the regulator member’s choice (a pure Θ^2 piece folds into the G_{cosmo} renormalization; other dependencies give other effective equations of state). **The family lives here.**
3. **The perturbation sector** $\delta\mathcal{W}$ — what the CMB and structure formation actually probe; §6.

The requirements map — the five filters. For the sector to discharge the inherited debts, qualitatively (this table *is* the filter set referenced throughout):

Probe	Requirement on the Θ -sector
BBN	total H^2 -tracking contribution inside the G_{cosmo}/G_N bound
CMB peaks	a component of the right abundance clustering like pressureless dust ($w \approx 0$, $c_s \approx 0$) at recombination
Clusters	the same dust-like component supplying the mass MOND alone cannot
Late times	the Λ -order constant (<i>automatic</i>)
Čerenkov / stability	propagating modes fast enough along cosmic-ray paths; FRW perturbations healthy

The hard row is the CMB’s: it is where MOND-class theories historically die.

6. The Clustering Dial: the Soft Spot Is the Doorway

The paper’s central structural result, boxed:

The One-Dial Result. The condition for dust-like clustering is $c_s \approx 0$ — and a near-zero sound speed at small A is exactly the near-vacuum behavior flagged as the deep-IR soft spot. **The same near-vacuum behavior that looked like a vulnerability is exactly what the CMB requires for dust-like clustering: one dial controls both.** What a dark-matter-like cosmological component requires — a fluid that does not free-stream, does not oscillate acoustically, and falls into potential wells like dust — is what the slowing, weakly-propagating near-vacuum scalar *is*. **The degeneracy cure and the clusters/CMB debt are the same dial.**

The trade the regulator member must thread:

- **Too stiff** ($c_s \sim 1$ near vacuum): degeneracy over-cured, Čerenkov trivially safe — but nothing clusters; the CMB debt stands in full.
- **Too soft** ($c_s = 0$ exactly): maximal dust-like clustering — but caustics form (the known pathology of pressureless scalar dust) and strong coupling returns.

- **The window** (small but nonzero c_s in the cosmological regime): dust-like clustering on CMB/cluster scales with caustics regulated — the trade the *imperfect-fluid* refinements of the class navigate.

So “choosing the regulator member” is not bookkeeping: **it is the single choice that simultaneously sets the degeneracy cure, the Čerenkov compliance, and the theory’s answer to clusters and the CMB.** One dial, three previously-separate ledger items.

6.1 Class mechanisms (directions, not verifications)

The class realizes both ends of the dial [I]: mimetic dark matter — a constrained unit-gradient scalar, structurally a constrained khronon, contributing *exact* pressureless dust at $c_s = 0$, with the known caustic issues that motivated the imperfect-fluid refinements — and **dark matter as an integration constant**, projectable Hořava’s global-only Hamiltonian constraint leaving a dust component. The honesty flag on the latter, repeated: that is the *neighboring* branch of the family — a proof-of-concept that foliation gravity can manufacture dust, not automatically ED’s mechanism.

A consonance, noted carefully. In both mechanisms the dust *abundance* is an initial condition or integration constant — not predicted. Under ED’s own methodology this is not a defect but a classification: **the existence of the dust channel would be form-level structure; the observed abundance would be value-inherited** — the same form/value split the corpus applies to every constant. The class’s “weakness” is ED’s standing epistemic posture.

The bridge to what follows: **the same single scale that governs the clustering dial also governs the vacuum term** — the cosmological face of the khronon has only one number to offer, and §7 follows that number to the cosmological constant.

7. One Λ : the Boundary’s Vacuum Imprint, at Two Levels

The corpus carried two Λ -mechanisms, flagged in KM-I §7.4 as faces to be reconciled. Read from source:

- **The V1 face (Paper_038.5; substrate level).** Λ as the V1 kernel’s **UV-finite** vacuum integral (the form factor cuts off at ℓ_P ; no mode-tower divergence) over the cosmological boundary $R_H = c/H_0$; verdict M3 (form-IDENTIFIED + value-INHERITED + IC-INHERITED), the value landing at $\rho_\Lambda = (3/8\pi)H_0^2 M_P^2 \sim H_0^2 c^2/G$. Its own honesty stands untouched: the naïve cutoff estimate failed by $\sim 10^{60}$ and is preserved as a record; the matter-sector hierarchy is open, reduced to Route-A closure.
- **The khronon face (KM-I §7.4 / §5 here; EFT level).** The foliation sector’s vacuum term, $\rho_\Lambda \sim a_0^2 c^2/G \sim H_0^2 c^2/G$, by the single-scale argument. Form-tier; $\mathcal{W}_0$ undetermined.

The scale identity is documentary — both faces carry $H_0^2 c^2/G$ by each paper’s own statement — **and automatic**: both are projections of the *one SCBU boundary*, the **kernel projection** (V1’s integral over R_H) and the **foliation projection** (the khronon’s vacuum term at its H_0 background). Agreement is what one boundary under two descriptions must produce; it is common origin, not coincidence, and not double-counting.

The level diagnosis and the thesis. The faces are not peers in one action: one is a *substrate-level mechanism*, the other an *EFT-level constant*. By standard effective-theory logic — low-energy constants *encode* the underlying theory rather than adding to it (the chiral-perturbation-theory pattern) —

One Λ , two descriptions: the khronon’s vacuum constant $\mathcal{W}_0$ is the **EFT encoding of the substrate-level V1 backreaction integral**. Λ is the SCBU boundary’s vacuum imprint — computed at the substrate by the V1 kernel (Paper_038.5), carried in the effective theory by the khronon (this arc).

7.1 The matching, performed

The matching is carried out to its honest limit, under one governing rule: **matching transports verdict tiers, it does not upgrade them** — a coefficient fixed by matching to a value-inherited result is itself value-inherited; moving a number between description levels cannot launder inheritance into derivation.

- **The condition.** Friedmann-level equality on the shared background: $\rho_\Lambda^{\text{EFT}}(\mathcal{W}_0) = \rho_\Lambda^{\text{substrate}}$. Under matching, the EFT constant is *defined* by the substrate computation — **double-counting is procedurally dissolved**, not merely argued away.
- **The scaling identity (derived, parameter-free).** Both faces scale exactly as $H_0^2 c^2 / G$: the substrate via the R_H domain and the Route-A memory scale $\ell_{V5}(H_0)$; the EFT via a_0^2 . No other scale survives on either side.
- **The sign check (derived; it could have failed).** The V1 vacuum density $\rho_{\text{vac},V1} = \frac{1}{2} \int (d^3k / (2\pi)^3) \omega_k |\hat{K}_{V1}|^2$ is **positive-definite by construction**, so under 038.5’s stated proportionality $\Lambda > 0$ — the observed de Sitter sign — fixing $\mathcal{W}_0 < 0$. (Consistent in flavor with KM-I’s IR Cancellation: both foliation-sector IR constants land negative in this convention.) Caveat named: this is the density-level pass; the full curvature-coupling chain belongs to the ab-initio task.
- **The coefficient transcription (D-via-I).** With $\Lambda_{\text{obs}} = 3\Omega_\Lambda H_0^2 / c^2$ and the EFT term $(c^4 / 16\pi G)(a_0^2 / c^4)\mathcal{W}_0 \equiv -(c^4 / 16\pi G) 2\Lambda$:

$$\mathcal{W}_0 = -24\pi^2 \Omega_\Lambda \approx -1.6 \times 10^2$$

against a_0^2 — equivalently $6\Omega_\Lambda \approx 4.1$ **against the raw background scale** $(cH_0)^2$, **genuinely order-unity** (the apparent largeness is the $(2\pi)^2$ of a_0 ’s definition, not a hierarchy; no naturalness claim is made either way). **The khronon EFT thereby loses its last free cosmological constant:** the regulator family retains only its Θ -*shape*; its endpoint is pinned — at the inherited tier, per the governing rule.

The falsifiable edge, upgraded to a number. What remains open is the **ab-initio substrate evaluation** of the V1 boundary integral (Paper_038.5’s Route-A frontier, unchanged in difficulty, including the coupling-chain sign). That computation, when performed, **must land on** $\mathcal{W}_0 = -24\pi^2 \Omega_\Lambda$ — wrong sign, or wrong beyond the computation’s honest error, breaks the one- Λ thesis and exposes an internal tension between the substrate and EFT faces. The thesis has a number to miss.

The clamps: no value of Λ is derived (the transcription inherits 038.5’s D-via-I closure — the tier-transport rule forbids reading it otherwise); the matter-sector cosmological-constant problem is untouched (the khronon face polices only the foliation sector’s own term); Ω_Λ is observational; the $\mathcal{O}(1)$ of the *substrate* integral is still uncomputed; the Θ_{ED} think-don’t-chase discipline stands.

8. Position — the Khronon’s Portfolio

With this paper the khronon’s roles close into one shape:

One field, one scale (H_0), five roles: the **arrow of time** (GR-II — its origin), the **Einstein sector** at high accelerations (GR-I — enforced by commitment-linearity), the **MOND sector** below a_0 (KM-I — the deep field), the **dark-fluid clustering dial** on FRW (§5–§6 — the clusters/CMB sector), and the **Λ -order vacuum term** (§7 — the boundary’s imprint). Every role is anchored to the same number because the field has only one number to offer.

Relations: this paper **completes the cosmological half** KM-I opened (its §7.4 flag is resolved; its soft spot is re-classified as the clustering dial); it **unifies with, and does not displace, Paper_038.5** (one Λ , two levels — 038.5 remains the substrate mechanism and keeps its verdict tier); and it gives **SCBU** its second dynamical statement: the boundary $R_H = c/H_0$ now has both a carrier field (KM-I) and a vacuum imprint computed at one level and carried at the other. Throughout, the **Orthogonality Theorem** (§4) is what licenses this portfolio reading: the five roles coexist in one field *because* the cosmological roles are sequestered from the tested galactic ones — the portfolio is unified at the level of the field, independent at the level of the tests.

9. The Wedge — Empirical Content

1. **The CMB as the dial’s judge.** The sharpest future test of the whole construction: the chosen regulator member must produce dust-like clustering of the right abundance at recombination, with the *same* function then fixed for clusters, Čerenkov, and BBN. A function with five filters has nowhere to hide.
 2. **The galactic sector is already locked.** By the Orthogonality Theorem (§4), nothing the cosmological data select can retroactively adjust the tested galactic physics — so any future CMB success cannot be a retrofit of the MOND sector, and any failure cannot be repaired from it. The two sectors test each other’s honesty.
 3. **The ab-initio V1 integral** — now carrying a *numeric* target (§7.1): it must land on $\mathcal{W}_0 = -24\pi^2\Omega_\Lambda$, with the sign already checked at the density level. A derivation with a number to miss.
 4. **Standing wedges inherited:** wide binaries near a_0 and the dynamical τ -sector (KM-I); the scalar GW polarization (GR-II).
 5. **Not claimed as wedges:** a Λ value; a CMB fit; the dark-matter abundance (value-inherited if the channel exists).
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10. Falsification Criteria

10.1 The window closes

Falsifier sentence: *A demonstration that no $\mathcal{W}(A^2, \Theta)$ member simultaneously satisfies the five filters — BBN, dust-like clustering at recombination, caustic regulation, Čerenkov, FRW stability — while remaining Θ -trivial in statics would falsify the §6 resolution: the clusters/CMB debt would return to the ledger as undischageable within this construction.*

10.2 The CMB rejects every dial setting

Falsifier sentence: *CMB data requiring properties of the dust-like component that no $\rho_{\mathcal{W}}(H)$ fluid with near-zero sound speed can produce (given the galactic-side sector already fixed by KM-I) would falsify the khronon’s cosmological face as the address for the MOND-class debts.*

10.3 The Λ faces disagree

Falsifier sentence: *Performance of the ab-initio substrate evaluation of the V1 boundary integral yielding a result inconsistent with $\mathcal{W}_0 = -24\pi^2\Omega_\Lambda$ — wrong sign (already checked at the density level, §7.1) or wrong beyond the computation’s honest error — would falsify the one- Λ thesis and expose an internal tension between Paper_038.5 and this arc.*

10.4 BBN

Falsifier sentence: *A required regulator member whose H^2 -tracking contribution violates the G_{cosmo}/G_N BBN bound would falsify that member — and, if all clustering-capable members so violate, the construction’s cosmological face.*

10.5 Sequestration breached

Falsifier sentence: *Any demonstration that a Θ -sector choice alters a static-sector observable (the modified Poisson equation, lensing, screening) would contradict the orthogonality theorem — a kinematic result — and would indicate the construction’s congruence analysis is wrong at root.*

10.6 Standing fronts (inherited)

Falsifier sentence: *KM-I’s falsification criteria (lensing slip; the Combination-Rule embedding; wide binaries; the interpolation family; α_1, α_2 — the last now resolved favorably in GR-IV, $\alpha_2 = 0$ exactly and α_1 safe by ≥ 70 orders) carry over unchanged; this paper adds no protection against any of them.*

11. Position Statement

The khronon’s cosmological face is structured, sequestered, and sits exactly where the data will judge it. The deep-vacuum degeneracy is resolved without retrofit — the degenerate point is outside ED’s state space (no Minkowski vacuum: the tick never halts), and the pure-acceleration form was a truncation whose generic completion is healthy at the physical cosmological point. The completion’s Θ -dependence is an honest family, **provably invisible to**

everything the companion papers tested (the orthogonality theorem). On FRW that family is a dark fluid with an automatic Λ -order late-time constant; its perturbation sector sits — by the same near-vacuum behavior once flagged as the soft spot — at the dust-clustering edge the CMB demands: **the degeneracy cure and the clusters/CMB debt are one dial**, with the class realizing both ends and the abundance entering as a value-inherited initial condition. And the corpus has **one Λ** : the khronon’s vacuum term is the EFT face of Paper_038.5’s substrate-level V1 integral over the one SCBU boundary — the matching *performed* (scaling exact; sign passing; $\mathcal{W}_0 = -24\pi^2\Omega_\Lambda$ pinned at the inherited tier, the EFT’s last free cosmological constant gone), the ab-initio integral open with a numeric target, the matter-sector problem left where it was.

What this paper claims. The structure: the state-space exclusion, the truncation diagnosis, the orthogonality theorem, the dark-fluid form with its requirements map, the dial identification, and the one- Λ thesis with its named computation.

What this paper does not claim. A chosen regulator member; a computed spectrum; a discharged cluster or CMB debt; a derived Λ value or abundance; a dust mechanism shown to live in ED’s own branch; or any change to the matter-sector cosmological-constant problem.

Series context. Paper KM-II of the post-pivot curvature-emergence line — the cosmological companion to KM-I, closing the structural analysis of the unification. What remains is model-building (set the dial), one named derivation (the Λ matching), the guarded primitives question, and the sky.

KM-I said: one scalar, one scale, one gravity. KM-II completes it: **one field, one scale, five roles — and the data as judge.**

Appendix: Cross-References, Glossary, Reader Map

Cross-references

- **Paper_087:** position paper. — **Papers_028/029:** the boundary $R_H = c/H_0$; a_0 .
- **Paper_038.5:** Λ as the V1 vacuum integral (the substrate face; unified with in §7).
- **Paper GR-I / GR-II / KM-I:** the metric and lapse; the class and the khronon; the galactic unification. (*inherited*)
- **SCBU:** the one boundary — two projections (§7), one carrier (KM-I).

Glossary

- **Regulator family.** The Θ -dependence of the general foliation sector $\mathcal{W}(A^2, \Theta)$; unfixed, sequestered, five-filtered.
- **Orthogonality theorem.** $\theta \equiv 0$ in statics $\implies$ the regulator family is invisible to every tested static result. No leak-back; no protection.
- **Dark fluid $\rho_{\mathcal{W}}(H)$.** The Θ -sector on FRW: a fluid whose density is a function of the expansion rate; Λ -order constant at late times.
- **The clustering dial.** The near-vacuum sound speed: too stiff = no clustering; too soft = caustics; the window = dust-like clustering with regulation. The same dial that cures the degeneracy.

- **The soft spot is the doorway.** The R3-flagged near-vacuum slow-mode behavior is the dust-clustering condition the CMB demands.
- **One- Λ thesis.** $\mathcal{W}_0$ as the EFT encoding of Paper_038.5's substrate-level V1 boundary integral: one constant, two descriptions. Matching performed (§7.1): scaling exact, sign passing, $\mathcal{W}_0 = -24\pi^2\Omega_\Lambda$ at D-via-I; the ab-initio integral open with that numeric target.
- **Tier-transport rule.** Matching transports verdict tiers; it does not upgrade them — a coefficient fixed by matching to a value-inherited result is itself value-inherited.
- **Mimetic dust / integration-constant dust.** Class proofs-of-concept at the dial's $c_s = 0$ end and in the neighboring projectable branch, respectively. Directions, not verifications.
- **No Minkowski vacuum.** ED's substrate forbids $(A, \theta) = (0, 0)$: the tick never halts. The degenerate point is excluded by the ontology.

Reader map and open work

Where to look next. - *The galactic half (the deep field, lensing, screening):* KM-I. — *The field and the class:* GR-II. — *The metric:* GR-I. - *The substrate face of Λ :* Paper_038.5. — *The boundary:* SCBU / Papers_028–029.

Open work (declared): the **ab-initio V1 boundary integral** — the cosmology thread's remaining derivation, now carrying the §7.1 numeric target ($\mathcal{W}_0 = -24\pi^2\Omega_\Lambda$); **setting the dial** — the minimal $\mathcal{W}(\Theta)$ parametrization under the five filters, with its endpoint now pinned (model-building); the **CMB/cluster confrontation** of the chosen member (observation); the guarded **primitives-origin** of the deep-IR branch; and KM-I's standing galactic fronts (wide binaries; the τ -sector; α_1, α_2).

End of Paper KM-II.

Post-pivot curvature-emergence line, the cosmological companion to KM-I. The deep-vacuum degeneracy is excluded by ED's own ontology (no Minkowski vacuum — the tick never halts) and diagnosed as a truncation artifact; the regulator family is sequestered by the orthogonality theorem (provably invisible to every tested result); the khronon's FRW face is a dark fluid with an automatic Λ -order constant whose perturbation sector sits at the dust-clustering edge — the soft spot is the doorway, and the degeneracy cure and the clusters/CMB debt are one dial under five filters. The corpus has one Λ : the khronon's vacuum term is the EFT encoding of the V1 boundary integral, with the matching performed — scaling exact, sign passing, $\mathcal{W}_0 = -24\pi^2\Omega_\Lambda$ pinned at the inherited tier (matching transports tiers, it does not upgrade them) — and the ab-initio integral open with that number as its target. One field, one scale, five roles — arrow, Einstein sector, MOND sector, clustering dial, vacuum term — and the data as judge. Clusters/CMB not discharged; no Λ value derived; no dial member chosen.